Faculty 4: Computer Science

Balancing Data Utility and Privacy in Medical Synthetic Data Generation: A Comparative Study of GAN-based and Large Language Model Approaches

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Abstract

The release of medical data for research is constrained by privacy regulations such as HIPAA and GDPR, while classical anonymization techniques have repeatedly failed under linkage attacks. Synthetic data generation offers a promising alternative, but the trade-off between data utility and privacy preservation remains poorly characterised — particularly when comparing fundamentally different generative paradigms. This thesis presents a unified empirical comparison of four approaches on the *UCI Diabetes 130-US Hospitals* dataset: HealthGAN (WGAN-GP, no formal differential privacy), PATE-GAN (GAN with (ε, δ)-DP), Pythia-LoRA (a large language model fine-tuned on serialised tabular records, no formal DP), and Pythia-DP (the same pipeline trained with DP-SGD via Opacus). Each generator is evaluated under two pillars. The first is *utility*, defined to encompass both *statistical similarity* to the real data (per-column distributions, correlation difference, Kolmogorov–Smirnov, PCA manifold overlap) and *predictive performance* of downstream models trained on the synthetic data (TSTR classifier performance, feature-importance preservation, Cox proportional-hazards concordance). The second is *privacy* (nearest-neighbour distance, NN adversarial accuracy, membership inference, attribute inference, outlier linkability). The thesis contributes (i) a reproducible evaluation pipeline, (ii) a head-to-head benchmark spanning the GAN–LLM and DP–non-DP axes, and (iii) empirical evidence on the marginal cost of formal differential privacy within each generative family.

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Chapter 1

Introduction

1.1 Background and Motivation

The exponential growth of digital health records, together with rapid advances in artificial intelligence (AI), offers significant potential to transform healthcare delivery. At the same time, robust regulatory frameworks such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) play a crucial role in safeguarding patient confidentiality and ensuring responsible data use. These protections are essential for maintaining public trust in digital health systems, but they also require researchers to explore innovative methods that enable data-driven innovation while upholding proper privacy standards. Conventional anonymisation methods have repeatedly proven insufficient to guarantee privacy in high-dimensional health data, as re-identification and linkage attacks remain viable [1, 2].

Synthetic data generation (SDG) has emerged as a promising alternative, enabling the creation of artificial datasets that retain the statistical properties of the original while mitigating privacy risks. Among the most prominent approaches are *Generative Adversarial Networks* (GANs) and *Large Language Models* (LLMs). GANs have demonstrated success in structured medical datasets and imaging data, while transformer-based LLMs offer advantages in modelling heterogeneous and unstructured records [3]. Generative Adversarial Networks work by training two neural networks — a generator that creates synthetic samples and a discriminator that tries to distinguish them from real data — until the generator learns to produce highly realistic outputs. Large Language Models operate using transformer architectures that leverage self-attention to learn long-range patterns in sequences, allowing them to generate coherent text or structured records. While GANs focus on matching data distributions through adversarial learning, LLMs generate data by predicting the next most probable token based on context.

The need for privacy-preserving data generation in healthcare is not only a technical challenge but a foundational concern of medical research. The scarcity of openly accessible, high-quality clinical data constrains innovation in diagnosis, treatment planning, and epidemiological modelling. Regulatory restrictions and ethical imperatives make direct sharing of identifiable records infeasible. Synthetic data presents a pathway to alleviate this bottleneck by allowing the distribution of data without direct disclosure of personal identifiers, thus accelerating AI model development and validation.

1.2 Problem Statement: The Utility–Privacy Tension

The adoption of synthetic data hinges on balancing two competing objectives:

1. **Utility:** the synthetic dataset must preserve the statistical and structural characteristics necessary for downstream analytical tasks.
2. **Privacy:** the dataset must provide strong resistance to adversarial attacks, such as membership inference and attribute disclosure.

While GANs and LLMs each offer unique strengths, there is limited comparative work evaluating their effectiveness under a unified privacy–utility framework, particularly for heterogeneous healthcare datasets.

1.3 Research Questions

This thesis addresses the following research questions:

RQ1. Which methods and metrics are used to evaluate privacy risks and utility (including predictive performance and distributional resemblance) in synthetic medical data?

RQ2. What are the trade-offs in utility and privacy in GAN- and LLM-based synthesised data generation models?

RQ3. Which synthetic data generation approach, based on GANs or Large Language Models, achieves a more effective balance between data utility and privacy preservation when applied to medical datasets?

1.4 Contributions

The work in this thesis makes the following contributions:

- A unified, reproducible evaluation pipeline for synthetic medical tabular data spanning two pillars — *utility* (encompassing both statistical similarity to the real data and predictive performance of downstream models) and *privacy* — with seventeen quantitative metrics in total.
- A head-to-head empirical comparison of four generators on the *Diabetes 130-US Hospitals* dataset [4]: HealthGAN (WGAN-GP) [5], PATE-GAN ((ε, δ)-DP) [6], Pythia-LoRA [7, 8, 3], and Pythia-DP-LoRA (DP-SGD via Opacus [9, 10]).
- Empirical evidence on the *marginal cost of formal differential privacy* within both the GAN family (HealthGAN $\rightarrow$ PATE-GAN at $\varepsilon = 5.0$) and the LLM family (Pythia $\rightarrow$ Pythia-DP at $\varepsilon = 5.0$). The Pythia-DP variant constitutes a scope extension beyond the originally submitted proposal.
- Practical recommendations for medical data stewards, indicating when a non-DP GAN suffices, when formal DP via PATE-GAN is warranted, and where LLM-based synthesis is competitive given privacy budgets and governance constraints.

1.5 Thesis Outline

The remainder of this thesis is structured as follows. **Chapter 2** surveys the background and related work on tabular synthetic data generation, GAN- and LLM-based approaches, differential privacy, and existing evaluation frameworks. **Chapter 3** describes the *Diabetes 130-US Hospitals* benchmark and the feature-engineering pipeline used to prepare it for synthesis. **Chapter 4** details the four generators compared in this work — HealthGAN, PATE-GAN, Pythia-LoRA, and Pythia-DP-LoRA — including architectures, training protocols, and the differentially private mechanisms employed. **Chapter 5** defines the unified two-pillar evaluation framework: utility (statistical similarity and predictive performance) and privacy. **Chapter 6** reports the empirical findings, structured around the three research questions. **Chapter 7** interprets the results, discusses strengths and weaknesses of each generator, and acknowledges threats to validity. Finally, **Chapter 8** summarises the contributions, returns directly to RQ1–RQ3, and outlines avenues for future work.

Chapter 2

Background and Related Work

2.1 Medical Tabular Data and Privacy Regulations

Due to the growing volume of digital health data, a variety of anonymisation techniques have been developed to address privacy concerns. Methods such as generalisation, suppression, distortion, swapping, and masking have been introduced [11], but most share a fundamental limitation: preserving utility under aggressive de-identification is difficult. Stadler, Oprisanu, and Troncoso [2] note that, although synthetic data often outperforms naïve anonymisation, a stable trade-off between preserving meaningful information and safeguarding individual privacy remains elusive.

2.2 Synthetic Data Generation: A Taxonomy

2.3 GAN-based Approaches for Tabular Data

2.3.1 Foundations: GAN, WGAN, WGAN-GP

2.3.2 Healthcare-specific Variants: medGAN, ADS-GAN, HealthGAN

ADS-GAN proposes a utility-first, non-DP GAN that adds an *identifiability* regulariser — a weighted distance that discourages generators from producing near-duplicates of real patients, with rare features receiving higher penalties [12]. The paper reports strong realism and competitive TSTR performance relative to baselines, while providing an interpretable privacy knob distinct from formal DP.

HealthGAN was introduced as a response to the limitations of medGAN, which could only generate binary diagnosis codes and often produced unrealistic distributions, making it unsuitable for mixed-type EHR data [5, 13, 14]. To address these shortcomings, the model combines medGAN's design with the WGAN-GP framework, leveraging the Wasserstein distance and a tailored data transformation pipeline to support continuous and categorical variables commonly found in datasets such as MIMIC-III [5]. Its architecture employs large batch sizes to ensure that outliers and rare clinical values are reliably captured, and incorporates a bottleneck to reduce memorisation and maintain a compact model footprint, facilitating secure model export outside the training environment. A key contribution of HealthGAN is the introduction of *Nearest-Neighbour Adversarial Accuracy* (AA), a metric suite that enables joint assessment of resemblance and privacy and has been recognised in systematic reviews as an important development in synthetic health data evaluation [1]. Empirical results show that HealthGAN was the only model in a six-method benchmark that maintained privacy while supporting model export, achieving competitive utility and resemblance relative to established baselines [5].

2.3.3 PATE-GAN

PATE-GAN adapts the Private Aggregation of Teacher Ensembles (PATE) framework to generative modelling by training multiple teacher discriminators on disjoint data partitions and aggregating their predictions through a differentially-private noisy mechanism, thereby providing formal (ε, δ)-DP guarantees while avoiding direct exposure of private gradients [6, 15]. This provides a tunable privacy budget with an explicit accountant, which aligns with survey guidance to report both formal guarantees and empirical privacy audits [16, 1]. In domains where non-DP synthetic data has been shown to leave outliers vulnerable [2], and where DP graphical approaches (e.g., PrivSyn [17]) trade utility for stronger privacy, PATE-GAN offers a principled middle ground with controllable ε.

2.4 LLM-based Approaches for Tabular Data

2.4.1 Tabula and the Row-as-Text Paradigm

Miletic and Sariyar [3] use a framework called *Tabula* that treats each row as text and shows that Pythia, a family of decoder-only language models, can match or surpass a strong tabular GAN (e.g., CTGAN [18]) on TSTR utility, with gains increasing in model size. The study focuses on *utility* rather than privacy — a gap explicitly addressed by this thesis.

2.4.2 The Pythia Model Family

2.4.3 LoRA Fine-tuning for Parameter-Efficient Adaptation

2.4.4 DP-Augmented LLM Synthesis

SafeSynthDP demonstrates that LLM-driven synthesis can be paired with differential privacy (e.g., Laplace and Gaussian mechanisms) to trace explicit ε–utility frontiers while maintaining competitive downstream accuracy and resisting membership inference [19]. Hyperparameters are adjusted along the privacy and accuracy axes respectively. This work positions DP-augmented LLM synthesis — the regime occupied by Pythia-DP in this thesis — as an active and largely unexplored frontier.

2.5 Differential Privacy

2.5.1 (ε, δ)-Differential Privacy and Composition

Differential privacy provides a formal mathematical framework for quantifying the privacy guarantees of randomised algorithms, and forms the theoretical foundation for the differentially-private generators evaluated in this thesis. A randomised mechanism $\mathcal{M}$ is said to satisfy (ε, δ)-differential privacy if, for every pair of neighbouring datasets D, D' that differ in a single record, and for every measurable output subset $S \subseteq \mathrm{Range}(\mathcal{M})$:

$$\Pr[\mathcal{M}(D) \in S] \;\leq\; e^{\varepsilon} \Pr[\mathcal{M}(D') \in S] + \delta. \tag{2.1}$$

The parameter ε denotes the privacy budget: smaller values impose stricter limits on the multiplicative ratio between output distributions, with $\varepsilon = 0$ corresponding to statistical indistinguishability. The relaxation term δ permits a small probability of catastrophic failure and is conventionally chosen to be substantially smaller than the inverse dataset size, with $\delta = 10^{-5}$ representing a standard convention for datasets of order 10^5 records [20].

The privacy guarantee of a deterministic query f is converted into a randomised mechanism by adding noise calibrated to the function's *global sensitivity*, defined as the maximum change in f over all pairs of neighbouring datasets. The Laplace and Gaussian mechanisms add noise proportional to this sensitivity divided by ε to satisfy pure ε-DP and approximate (ε, δ)-DP respectively [21, 20].

Iterative procedures such as the differentially-private optimisation algorithms employed in this thesis perform many noisy queries against the private dataset, requiring composition rules that quantify how the cumulative privacy cost grows. Basic sequential composition states that k mechanisms, each $(\varepsilon, 0)$-differentially private, yield a $(k\varepsilon, 0)$-private composite, but this linear bound is loose for practical training regimes that invoke thousands of gradient computations. The advanced composition theorem of Dwork and Roth [20] tightens the bound to approximately $O(\sqrt{k \log(1/\delta)}\,\varepsilon)$ for k adaptive queries; even this remains insufficient for the repeated Gaussian noise injections of DP-SGD, motivating the tighter accountants introduced in §2.5.4.

2.5.2 DP-SGD and Opacus

The Differentially-Private Stochastic Gradient Descent algorithm of Abadi et al. [10] extends standard mini-batch SGD with two privacy-preserving operations applied at every training step. Given a model with parameters θ, loss $\mathcal{L}$, clipping threshold C, noise multiplier σ, and a Poisson-sampled mini-batch B in which each training example is included independently with probability p, the algorithm computes per-example gradients $g_i = \nabla_\theta \mathcal{L}(\theta, x_i)$ and clips each to a bounded L2 norm:

$$\bar{g}_i = \frac{g_i}{\max\big(1,\ \|g_i\|_2 / C\big)}. \tag{2.2}$$

The clipped per-example gradients are then summed, perturbed by isotropic Gaussian noise, and averaged to obtain the noisy gradient used for the parameter update:

$$\tilde{g} \;=\; \frac{1}{|B|}\left(\sum_{i\in B} \bar{g}_i \;+\; \mathcal{N}\big(\mathbf{0},\, \sigma^2 C^2\,\mathbf{I}\big)\right), \qquad \theta \;\leftarrow\; \theta - \eta\,\tilde{g}. \tag{2.3}$$

Per-example clipping bounds the contribution any single training record can make to a gradient update, capping the L2 sensitivity of the gradient sum at C and rendering the addition of Gaussian noise with standard deviation σC sufficient to provide a formal privacy guarantee. The noise multiplier σ controls the privacy–utility trade-off: larger values yield smaller privacy budgets at the cost of noisier gradients and slower convergence. The total privacy expenditure across T training steps is computed using a tight accountant (§2.5.4) rather than by naïvely composing the (ε, δ) guarantee of each step.

The Opacus library [9] provides an implementation of DP-SGD for the PyTorch ecosystem, exposing per-example gradient computation, gradient clipping, noise injection, and privacy accounting as extensions to standard training pipelines. Opacus is adopted in this thesis to train the Pythia-DP generator, with method-specific configuration details such as the chosen noise multiplier and target (ε, δ) deferred to Chapter 4.

2.5.3 The PATE Framework

The Private Aggregation of Teacher Ensembles (PATE) framework of Papernot et al. [15] offers an alternative route to differentially-private learning that avoids the explicit gradient perturbation of DP-SGD. The private training data is first partitioned into N disjoint subsets, and an independent *teacher* model is trained on each partition without any privacy mechanism. A separate *student* model is then trained on a corpus of unlabelled public data, with labels generated by aggregating the teachers' predictions through a noisy voting mechanism:

$$\hat{y}(x) \;=\; \arg\max_j \;\Big\{ n_j(x) \;+\; \mathrm{Lap}(1/\gamma) \Big\}, \tag{2.4}$$

where $n_j(x)$ denotes the number of teachers voting for class j on input x and $\mathrm{Lap}(1/\gamma)$ is Laplace noise with scale $1/\gamma$ controlling the privacy cost per query.

The disjointness of the partitions is essential to the construction: because any individual training record influences exactly one teacher, the sensitivity of the vote count to the substitution of a single record is bounded by one, allowing calibrated noise to confer a formal (ε, δ)-DP guarantee on the released labels. The student model is trained exclusively on these noisy aggregated labels and never accesses the private dataset, so the guarantee transfers to the trained student via the post-processing property of differential privacy. A favourable feature of the construction is that consensus among teachers consumes a smaller portion of the privacy budget than disagreement, allowing the framework to exploit easy queries efficiently. The Gaussian-noise variant of PATE admits tighter analysis under Rényi differential privacy (§2.5.4), and the application of the PATE construction to generative adversarial networks is described in §2.3.3.

2.5.4 Rényi DP and the Moments Accountant

The repeated application of the Gaussian mechanism in DP-SGD and the iterative noisy aggregation in PATE both require composition tools more refined than the basic and advanced composition theorems introduced in §2.5.1. Mironov [22] proposed Rényi Differential Privacy (RDP), a relaxation of standard differential privacy expressed in terms of the Rényi divergence of order $\alpha > 1$. A randomised mechanism $\mathcal{M}$ satisfies $(\alpha, \bar{\varepsilon})$-RDP if, for every pair of neighbouring datasets D, D':

$$D_\alpha\big(\mathcal{M}(D) \,\|\, \mathcal{M}(D')\big) \;\leq\; \bar{\varepsilon}, \tag{2.5}$$

where $D_\alpha(P\|Q)$ denotes the Rényi divergence of order α between distributions P and Q.

Rényi differential privacy admits a particularly simple composition rule: the sequential application of k mechanisms, each $(\alpha, \bar{\varepsilon})$-RDP, yields a $(\alpha, k\bar{\varepsilon})$-RDP composite. This linear growth at fixed α avoids the square-root inflation incurred by the advanced composition theorem of standard (ε, δ)-DP. An $(\alpha, \bar{\varepsilon})$-RDP mechanism may be converted to a standard (ε, δ)-DP guarantee via

$$\varepsilon \;=\; \bar{\varepsilon} \;+\; \frac{\log(1/\delta)}{\alpha - 1}, \tag{2.6}$$

for any $\delta \in (0, 1)$ [22]. In practice the accountant tracks the RDP cost at multiple orders α simultaneously during training and, upon conversion, selects the order that minimises the resulting ε at the chosen δ.

Rényi DP generalises the earlier *moments accountant* of Abadi et al. [10], which tracked the higher moments of the privacy-loss random variable to obtain tighter bounds for repeated Gaussian queries than basic composition would yield. The Gaussian mechanism, the subsampled Gaussian mechanism employed by DP-SGD with Poisson batching, and the Laplace mechanism employed by PATE all admit closed-form RDP bounds at every order α, making RDP the standard accounting framework for both algorithms.

2.6 Evaluation Frameworks for Synthetic Data

2.6.1 Utility – Statistical Similarity to the Real Data

Murtaza et al. [1] propose a taxonomy that distinguishes univariate metrics such as Kolmogorov–Smirnov (KS), Kullback–Leibler (KL) divergence, and moment-based comparisons from multivariate metrics including correlation matrix similarity, Maximum Mean Discrepancy, Jensen–Shannon divergence, and Wasserstein distance, all used to quantify how closely synthetic distributions match real data. Goncalves et al. [23] complement these distributional measures by combining KL divergence with pairwise correlation differences (PCD) to assess both marginal and joint-distribution fidelity in synthetic patient data. In the framework adopted here these metrics are treated as one branch of utility, since a synthetic dataset that fails to preserve the statistical structure of the real data cannot reasonably be expected to support downstream tasks.

2.6.2 Utility – Predictive Performance of Downstream Models

Hernandez et al. [16] survey tabular synthetic data generators and emphasise model-based utility evaluations, including Train-on-Synthetic Test-on-Real (TSTR) and related train–test regimes (TRTR/TRTS/TSTS), as central tools for assessing whether synthetic data supports realistic downstream inference. Reported metrics typically include accuracy, F1, and AUC. Wang, Zhang, and He [24], Yang et al. [25], Park et al. [26], and Beaulieu-Jones et al. [27] all employ such regimes; Emam, Mosquera, and Zheng [28] introduces Dimension-Wise Prediction (DWP) as a complementary multivariate utility metric. Together with the statistical-similarity metrics in the previous subsection, these predictive-performance metrics constitute the second branch of the utility pillar used throughout the thesis.

2.6.3 Privacy Metrics: Distance, Inference, and Linkability

Across the synthetic data literature, different studies employ diverse privacy metrics to quantify disclosure risks. The evaluation framework by Kaabachi et al. [29] measures membership inference and attribute inference for each patient record, together with privacy gain, which captures the reduction of re-identification probability when disclosing synthetic rather than real data. HealthGAN-based work evaluates privacy through Nearest-Neighbour Adversarial Accuracy (AA) and privacy loss, and also simulates membership inference attacks [13]. Hernandez et al. [16] summarise additional metrics, including identity/attribute disclosure, Distance to Closest Record (DCR), membership attack success rate, max real-to-synthetic similarity, formulation of differential privacy, privacy loss, and Maximum Mean Discrepancy (MMD). Murtaza et al. [1] systematically categorise privacy metrics into integrated DP mechanisms (DP, Rényi DP, moments accountant), embedded objectives such as ε-identifiability, ℓ-diversity, and probabilistic k-anonymity, and post-hoc metrics including nearest-neighbour similarity, NN attribute estimation, reproduction rate, outlier similarity, and distribution-based measures such as perplexity-distribution similarity. Several papers also simulate membership inference by matching real and synthetic records through distance thresholds (e.g., Hamming distance [23]) and evaluate attribute disclosure by testing whether unknown attributes can be inferred from nearest synthetic neighbours.

2.7 Research Gap and Positioning

Hernandez et al. [16], in their systematic review of tabular health data for synthetic data generation, conclude that no single method or metric is universally best, and survey resemblance, privacy, and utility metrics used by other authors. The thesis presented here directly addresses this gap by applying a unified evaluation pipeline to four representative generators spanning the GAN/LLM and DP/non-DP axes.

Chapter 3

Dataset and Preprocessing

3.1 The Diabetes 130-US Hospitals Dataset

The benchmark used throughout this thesis is the *UCI Diabetes 130-US Hospitals* dataset [4, 30]. It contains 101,766 inpatient encounters from diabetic patients treated across 130 U.S. hospitals between 1999 and 2008. Each record describes a hospital admission of 1–14 days and includes 47 mixed-type features covering demographics (age, gender, race), admission characteristics, laboratory test results, diagnoses, medication use, and prior inpatient, outpatient, and emergency visits. The data consists primarily of categorical and integer variables and represents detailed clinical, administrative, and treatment-related information collected during the patient's hospital stay. Missing values are present, and no predefined data splits are provided, enabling flexible train–validation–test partitioning. The binary target used throughout this work is `readmitted`: hospital readmissions of any kind are encoded as the positive class and "no readmission" as the negative class.

Scope. The proposal contemplated MIMIC-III as a secondary benchmark; the final thesis is restricted to the Diabetes 130-US Hospitals dataset to keep training budgets and evaluation runs manageable across four generators and seventeen metrics. Cross-dataset generalisation is discussed as future work in Section 8.3.

3.2 Exploratory Data Analysis

3.2.1 Schema and Data Types

The raw CSV contains 50 columns: 13 integer columns and 37 object columns when loaded directly with `pandas`. The integer fields include encounter and patient identifiers, utilisation counts, procedure counts, medication counts, diagnosis counts, and length of stay. Most remaining fields are categorical, including demographic variables, admission and discharge codes, ICD-9 diagnosis codes, laboratory-result indicators, medication-change indicators, and the readmission target. The preprocessing loader treats `?` as a missing value and then casts known count and identifier fields to numeric type while leaving the remaining columns as categorical variables.

3.2.2 Missingness Patterns

Missingness is concentrated in a small number of clinical and administrative fields. The raw columns with the most missing data are weight (96.86% missing), max-glucose-serum (94.75%), `A1Cresult` (83.28%), medical specialty (49.08%), and payer code (39.56%). Lower missingness is observed in `race` (2.23%), `diag_3` (1.40%), `diag_2` (0.35%), and `diag_1` (0.02%).

The pipeline removes `weight`, `payer_code`, and `medical_specialty` because their missingness is high and their values are not needed for the downstream synthetic-data comparison. `race` is also removed after resolving multiple encounters, avoiding use of a sensitive demographic field. `A1Cresult` and diagnosis fields are retained because their missingness can be represented or imputed after clinically motivated encoding.

3.2.3 Near-Zero-Variance Columns

Several medication variables are highly imbalanced in the raw dataset because almost all encounters record `No`. Fifteen medication fields have fewer than 1% non-mode values, including rare drugs such as `examide`, `citoglipton`, `troglitazone`, and several combination medications. The pipeline keeps these medication fields, but converts each from the original `No`, `Steady`, `Up`, and `Down` representation into a binary exposure indicator. This preserves whether a drug was used while reducing sparse categorical levels that would be difficult for tabular generators to model reliably.

3.2.4 Target Distribution and Class Imbalance

The raw target has three categories: `NO`, `>30`, and `<30`. In the full raw dataset, `NO` accounts for 54,864 encounters (53.91%), `>30` for 35,545 encounters (34.93%), and `<30` for 11,357 encounters (11.16%). For this thesis, the two readmission categories are collapsed into a single positive class because the main modelling task is whether any readmission occurred.

After multiple-encounter resolution and binary target encoding, the processed dataset contains 47,188 non-readmitted records (65.98%) and 24,330 readmitted records (34.02%). The stratified training split preserves this imbalance with 37,750 negatives and 19,464 positives. This class distribution is used as a baseline property that synthetic generators should reproduce.

3.3 Feature Engineering Pipeline

The preprocessing and feature-engineering code is implemented in the feature-engineering module and executed by the data-preprocessing script. The script loads the raw data, applies the engineering pipeline, writes the full preprocessed table, and then creates stratified train and test CSV files. The transformation order is fixed so that row-level deduplication and clinically meaningful grouping occur before imputation and label encoding.

3.3.1 Multiple-Encounter Resolution

The raw dataset contains repeated encounters for some patients. Although there are 101,766 encounter rows, there are 71,518 unique values of `patient_nbr`; 16,773 patients appear more than once, and the maximum number of encounters for a single patient is 40. To avoid patient-level leakage between train and test splits, `process_multiple_encounters` keeps one row per patient. It sorts by `patient_nbr` and `time_in_hospital`, retains the encounter with the longest hospital stay for each patient, and drops duplicate patient rows.

After this step, `patient_nbr` is removed because it is a direct patient identifier. The `race` column is also removed, reducing the use of sensitive demographic information in the modelling pipeline.

3.3.2 Age Bucket Consolidation

`consolidate_age` converts the original decade-based age buckets into ordered numeric values. The final experiments use a 1–10 ordinal encoding: `[0-10)` maps to 1, `[10-20)` maps to 2, and so on through `[90-100)`, which maps to 10. This preserves the natural ordering of age while avoiding a high-cardinality string representation.

An earlier preprocessing variant collapsed the ten decades into a three-level grouping (young / middle-aged / elderly) to reduce categorical sparsity. A direct ablation against the ten-bin encoding showed that the coarser grouping substantially degraded downstream synthetic-data utility for the LLM-based generator, while leaving the GAN baselines largely unaffected. The native ten-bin encoding is therefore retained for all experiments in this thesis. Quantitative results of the ablation are reported in Section 6.2.1.3.

3.3.3 Dropping High-Missingness Columns

This step removes three sparse fields: weight, payer code, and medical specialty. Weight is almost entirely missing, while payer code and medical specialty have substantial missingness and introduce many administrative categories. Dropping these fields reduces dimensionality and prevents the synthetic-data generators from spending capacity on sparse or poorly observed variables.

3.3.4 ICD-9 Diagnosis Grouping

`change_diag_columns` maps `diag_1`, `diag_2`, and `diag_3` from raw ICD-9 strings into broader diagnosis groups using `_icd9_to_group`. Codes beginning with 250 are mapped to `Diabetes`. Numeric ICD-9 ranges are grouped as circulatory, respiratory, digestive, genitourinary, neoplasm, musculoskeletal, and injury categories. External cause and supplementary codes beginning with `E` or `V`, missing values, and unrecognised codes are mapped to `Other`.

This step reduces thousands of possible diagnosis-code values into nine clinically interpretable categories. The categories are still categorical at this stage and are converted to integer labels later by the label-encoding step.

3.3.5 Binary Medication Encoding

`change_medication_columns` transforms 23 medication columns into binary indicators. For each medication, `No` is encoded as 0 and every other value, including `Steady`, `Up`, and `Down`, is encoded as 1. The encoded set includes the individual diabetes medications and combination therapies represented in the raw dataset, such as metformin, insulin, sulfonylureas, thiazolidinediones, and metformin-based combinations.

The binary representation emphasises whether a medication was prescribed or changed during the encounter, rather than attempting to model sparse dosage direction categories separately.

3.3.6 A1C Result Encoding

`encode_a1cresult` converts `A1Cresult` into an ordinal laboratory indicator. Missing A1C values are encoded as 0, `Norm` as 1, `>7` as 2, and `>8` as 3. This keeps the absence of an A1C test as an explicit value while preserving the severity ordering of observed test results.

3.3.7 Imputation Strategy

`impute_data` is applied after diagnosis grouping, medication encoding, and A1C encoding. Categorical columns are imputed with the most frequent observed value, while numeric columns are imputed with the median. The pipeline uses `SimpleImputer` for both operations. After imputation, the preprocessed dataset contains no missing values.

3.3.8 Categorical Label Encoding

`label_encode` converts all remaining object-type feature columns into integer labels using `LabelEncoder`. By default, the target column is excluded from this step so that it can be encoded with an explicit clinical mapping. This step is applied to categorical predictors such as gender, admission categories, discharge categories, glucose-serum result categories, diagnosis groups, medication-change indicators, and diabetes-medication indicators. The resulting preprocessed table contains only numeric columns.

3.3.9 Target Encoding

`target_encode` maps the original three-class readmission variable into a binary outcome. `NO` is encoded as 0, while both `>30` and `<30` are encoded as 1. The binary formulation treats any hospital readmission as the actionable positive outcome and is used consistently for model training, synthetic-data evaluation, and downstream metric computation.

3.4 Train/Test Split

The preprocessing script writes three CSV files under `thesis/data`: the full preprocessed dataset, the training split, and the test split. The full processed dataset contains 71,518 rows and 45 columns. The split is created with `train_test_split`, `test_size = 0.2`, and `random_state = 42`. The target column is passed as the stratification variable, preserving the binary readmission ratio across splits.

The final training set contains 57,214 rows and 45 columns, with 37,750 negative records and 19,464 positive records. The held-out test set contains 14,304 rows and 45 columns, with 9,438 negative records and 4,866 positive records. Both splits contain no missing values and no object-type columns after preprocessing.

Chapter 4

Synthetic Data Generation Methods

4.1 Selection Rationale

The four generators evaluated in this thesis are chosen to provide a 2×2 coverage of the design space: $\{\text{GAN}, \text{LLM}\}$ on the modelling axis, and $\{\text{non-DP}, \text{DP}\}$ on the privacy axis. Within each axis the choice is grounded in prior work:

- **HealthGAN (non-DP GAN baseline).** HealthGAN was designed specifically for mixed-type EHR tables, comes with a complete secure training-to-export workflow, and introduces nearest-neighbour Adversarial Accuracy (AA) to co-measure resemblance and privacy (TrainAA/TestAA and privacy loss), yielding competitive TSTR utility while keeping a small footprint [5, 13, 31].
- **PATE-GAN (DP GAN).** PATE-GAN enforces formal (ε, δ)-DP by decoupling privacy from adversarial training: multiple teacher discriminators are trained on disjoint partitions, their votes are noisily aggregated, and a student discriminator learns from these noisy labels so the generator inherits DP by post-processing [6, 15].
- **Pythia LLM (non-DP LLM baseline).** The benchmark study by Miletic and Sariyar [3] evaluates the Pythia family of decoder-only LLMs for tabular health data and shows they can match or exceed established tabular GANs. The authors introduce *Tabula*, a framework that treats each tabular row as a text sequence. Their evaluation focuses on utility rather than privacy.
- **Pythia-DP (DP LLM, scope extension).** Beyond the originally proposed three models, this thesis adds a differentially private variant of Pythia trained with DP-SGD via Opacus [9, 10], enabling a within-family DP-cost comparison parallel to the HealthGAN $\rightarrow$ PATE-GAN comparison.

4.2 HealthGAN: WGAN-GP Baseline

4.2.1 SDV Encoding and Decoding

4.2.2 WGAN-GP Architecture

4.2.3 Training Loss: Wasserstein + Gradient Penalty

4.2.4 Hyperparameters and Training Configuration

4.2.5 Variant Selection: samples_0 vs. samples_3 vs. samples_6

4.2.6 Rationale for Excluding a DP-Augmented HealthGAN Variant

A natural question is whether differential privacy could be applied directly to the HealthGAN training loop — for instance by replacing the Adam optimiser of the critic with a DP-SGD variant [10] via `tensorflow_privacy`. This thesis does not pursue that direction; the DP-GAN comparison is instead realised through the architecturally distinct PATE-GAN model (§4.3). Four technical considerations motivate this choice.

First, the gradient-penalty term that distinguishes WGAN-GP from weight-clipping WGAN [32, 33] is itself a function of real data, as it is computed over interpolated samples drawn between real and generated points. Its gradient cannot be cleanly clipped on a per-example basis as DP-SGD requires, so accommodating DP-SGD would necessitate dropping the gradient penalty and reverting to weight-clipping Wasserstein training, sacrificing the architectural property that motivated the WGAN-GP variant in the first place. Second, the training protocol applies five critic updates per generator update; because the critic is the sole component that directly processes real records, this $5{:}1$ ratio implies the privacy budget would be consumed five times faster than in a configuration performing one private gradient computation per logical batch, as is the case in the Pythia-DP setting (§4.5). Third, the injection of Gaussian noise into critic gradients — the core operation of DP-SGD — is well documented to exacerbate the mode collapse and non-convergence that already characterise GAN training on tabular health data, with empirical results in the DP-GAN literature [34, 6] demonstrating sharp utility cliffs at budgets of $\varepsilon \leq 10$. Fourth, the HealthGAN implementation relies on TensorFlow 1 compatibility mode with eager execution disabled, and the per-example gradient computation required by `tensorflow_privacy` is substantially slower under graph execution, rendering the $100{,}000$-epoch training schedule computationally prohibitive within the resource envelope of this project.

Beyond these technical obstacles, the experimental design already provides a controlled DP-GAN baseline through PATE-GAN, which attains formal (ε, δ)-DP through a fundamentally different mechanism — noisy aggregation of teacher discriminators [6, 15] — rather than by injecting noise into the critic's gradients. The HealthGAN $\rightarrow$ PATE-GAN pairing therefore isolates the cost of formal DP on the GAN side without confounding it with an unstable adaptation of an architecture that was not designed for per-example gradient clipping. HealthGAN is retained as the non-private GAN utility baseline.

4.3 PATE-GAN: Differentially Private GAN

4.3.1 Teacher–Student Framework

4.3.2 PATE Voting with the Laplace Mechanism

4.3.3 Privacy Accounting via the Moments Accountant

4.3.4 Architecture

4.3.5 Privacy Budget and Hyperparameters

4.4 Pythia: Non-DP LLM-based Generation

4.4.1 Base Model

4.4.2 Tabular-to-Text Serialisation

4.4.3 Schema Derivation

4.4.4 LoRA Fine-tuning

4.4.5 Class-Conditioned Generation

4.4.6 Schema-Aware Parsing, Coercion, and Retry

4.5 Pythia-DP: Differentially Private LLM

4.5.1 DP-SGD via Opacus

4.5.2 Per-Example Gradient Clipping

4.5.3 Effective Batch Size

4.5.4 Privacy Budget and Achieved Epsilon

4.5.5 Differences from the Non-DP Pipeline

4.6 Implementation Summary

Chapter 5

Evaluation Framework

5.1 Two Pillars: Utility and Privacy

This chapter defines a two-pillar evaluation framework: *utility* and *privacy*. The utility pillar is treated as a single concept with two complementary branches: *statistical similarity* between the synthetic and real datasets (Section 5.2.1) and *predictive performance* of downstream models trained on the synthetic data (Section 5.2.2). The privacy pillar is defined in Section 5.3 via instance-level distance and inference-attack metrics. Section 5.4 introduces the aggregation views that combine both pillars into a single comparative picture.

5.2 Utility Metrics

The utility pillar combines two complementary aspects. The first is whether the synthetic dataset *looks like* the real one in a distributional sense; the second is whether the synthetic dataset *acts like* the real one when used as training data for downstream predictive and survival models. The same set of synthetic records is evaluated under both views.

5.2.1 Statistical Similarity to the Real Data

5.2.1.1 Per-Column Distributional Match

5.2.1.2 Summary Statistics (Mean, Standard Deviation)

5.2.1.3 Pairwise Correlation Difference (PCD)

5.2.1.4 Kolmogorov–Smirnov Two-Sample Test

5.2.1.5 PCA-Based Manifold Overlap

5.2.2 Predictive Performance of Downstream Models

5.2.2.1 Train–Test Regimes: TRTR and TSTR

5.2.2.2 Classifier Suite

5.2.2.3 Performance Metrics

5.2.2.4 ROC Curve Analysis

5.2.2.5 Feature Importance Preservation

5.2.2.6 Cox Proportional Hazards Regression

5.2.2.7 Cox CI Overlap Analysis

5.3 Privacy Metrics

5.3.1 Nearest-Neighbour Distance

5.3.2 Nearest-Neighbour Distance Ratio (NNDR)

5.3.3 Nearest-Neighbour Adversarial Accuracy (NNAA)

5.3.4 Membership Inference Attack

5.3.5 Attribute Inference Attack

5.3.6 Clustering Proximity / Outlier Linkability

5.4 Comparative Aggregation

5.4.1 Privacy–Utility Tradeoff Scatter

5.4.2 Six-Dimension Radar Chart

5.4.3 Comprehensive Summary Table

Chapter 6

Results

6.1 RQ1 – Methods and Metrics for Evaluating Synthetic Medical Data

6.1.1 Operationalising the Two-Pillar Framework

6.1.2 Per-Pillar Diagnostic Value

6.1.3 Recommended Minimal Evaluation Suite

Answer to RQ1

6.2 RQ2 – Utility–Privacy Trade-offs Per Generator

6.2.1 Utility Results

6.2.1.1 Statistical Similarity

Per-Column Distributions and Marginals.

Pairwise Correlation Difference (PCD).

Kolmogorov–Smirnov Statistics.

PCA-Based Manifold Overlap.

6.2.1.2 Predictive Performance

TSTR Classifier Performance.

6.2.1.3 Ablation: Age Binning Granularity

The preprocessing pipeline (§3.3.2) retains the dataset's native ten-decade encoding of `age` after an earlier design that collapsed it into three coarse groups (young / middle-aged / elderly) was abandoned. Table 6.1 quantifies the impact on Train-on-Synthetic, Test-on-Real (TSTR) utility, evaluated on the real held-out test split ($n = 14{,}304$) with `readmitted` as the binary target.

The Pythia LLM generator improves substantially under the ten-bin encoding across all three classifiers: AUC-ROC rises by 0.044–0.067 and macro-F1 by 0.099–0.112. The Train-on-Real upper bound is essentially unchanged between the two preprocessing variants (Gradient-Boosting TRTR AUC: 0.7014 under the three-bin scheme vs. 0.7008 under the ten-bin scheme), confirming that the gain originates from improved synthetic generation rather than a shift in the real-data signal. Scaling Pythia from 70M to 1B parameters under the ten-bin encoding further raises Gradient-Boosting TSTR AUC to 0.6818, recovering $\approx 97.3\%$ of the TRTR upper bound.

Table 6.1: TSTR utility for the Pythia LLM generator under the discarded three-bin age encoding versus the retained ten-bin encoding. Test set: real held-out split ($n = 14{,}304$); target: `readmitted`. Δ reports the change from three-bin to ten-bin (Pythia-70M).

Classifier	Metric	3-bin (Pythia-70M)	10-bin (Pythia-70M)	Δ	10-bin (Pythia-1B)
Logistic Regression	AUC-ROC	0.6128	0.6570	+0.0442	0.6612
Logistic Regression	F1	0.4046	0.5038	+0.0992	0.4995
Random Forest	AUC-ROC	0.5917	0.6588	+0.0671	0.6682
Random Forest	F1	0.3843	0.4966	+0.1123	0.5048
Gradient Boosting	AUC-ROC	0.5981	0.6634	+0.0653	**0.6818**
Gradient Boosting	F1	0.3944	0.5005	+0.1061	0.5036
TRTR (real) Gradient-Boosting AUC		0.7014	0.7008	—	—

A plausible mechanistic explanation is that Pythia emits each field as a discrete token: under the three-bin scheme, three tokens absorbed the entire 0–100 year range, erasing conditional structure between age and clinically correlated features (`number_diagnoses`, `number_inpatient`, `insulin`). Restoring the ten-decade encoding provides a finer conditioning vocabulary, recovering age-specific co-occurrence patterns relevant to readmission prediction. The PATE-GAN and HealthGAN baselines did not exhibit a comparable sensitivity to this preprocessing choice, suggesting that categorical cardinality affects token-based LLM synthesisers more than GAN-based ones on tabular medical data. This observation is discussed further in Section 7.2.

ROC Curves (Gradient Boosting).

Feature Importance Stability.

Cox Proportional Hazards.

Cox CI Overlap.

6.2.2 Privacy Results

6.2.2.1 Nearest-Neighbour Distance and NNDR

6.2.2.2 Nearest-Neighbour Adversarial Accuracy

6.2.2.3 Membership Inference Attack

6.2.2.4 Attribute Inference Attack

6.2.2.5 Outlier Linkability

6.2.3 Within-Family DP Cost

6.2.3.1 HealthGAN → PATE-GAN

6.2.3.2 Pythia → Pythia-DP

Answer to RQ2

6.3 RQ3 – Which Approach Achieves Better Balance

6.3.1 Tradeoff Landscape

6.3.2 Six-Dimension Radar Comparison

6.3.3 Comprehensive Summary Table

6.3.4 Pareto Analysis: GAN Family vs. LLM Family

Answer to RQ3

Chapter 7

Discussion

7.1 Interpretation of Results per Research Question

7.2 GAN vs. LLM: Strengths, Weaknesses, and Use Cases

Sensitivity to Categorical Granularity. A practical asymmetry surfaced by the age-binning ablation (Section 6.2.1.3) is that token-based LLM synthesisers are more sensitive to upstream categorical-cardinality choices than GAN baselines. Collapsing the ten-decade `age` encoding into three coarse groups depressed Pythia's Gradient-Boosting TSTR AUC by ≈ 0.065 and macro-F1 by ≈ 0.106, while leaving HealthGAN and PATE-GAN approximately unchanged. The implication for healthcare data stewards: when an LLM-based generator is in scope, preprocessing decisions that flatten ordinal categorical variables for "simplicity" can silently forfeit a substantial fraction of recoverable downstream utility, and should therefore be validated against the dataset's native encoding before being committed to.

7.3 Practical Recommendations for Healthcare Data Stewards

7.4 Toward Standardisation of Synthetic-Data Evaluation

7.5 Limitations

7.5.1 Single-Dataset Scope

7.5.2 Pythia-70m Size Constraint

7.5.3 Single Privacy Budget ($\varepsilon = 5.0$)

7.5.4 Computational Cost

7.5.5 Scope Reductions Versus the Proposal

7.6 Threats to Validity

7.7 Ethical Considerations

Chapter 8

Conclusion and Future Work

8.1 Summary of Contributions

This thesis delivers a systematic and rigorously controlled comparison of GAN-based and LLM-based methods for privacy-preserving synthetic medical data generation. By evaluating HealthGAN, PATE-GAN, Pythia, and Pythia-DP under a unified experimental framework on the *Diabetes 130-US Hospitals* dataset [4], the study establishes a transparent basis for analysing the privacy–utility trade-off across fundamentally different generative paradigms. The integration of aligned training conditions, consistent preprocessing, and a 17-metric evaluation pipeline ensures that differences in performance reflect model characteristics rather than experimental variability.

The four concrete contributions of this work are:

1. **Unified evaluation pipeline.** A reproducible two-pillar framework — *utility* (split into statistical similarity to the real data and predictive performance of downstream models) and *privacy* — instantiated as 17 metrics applied identically to every generator under comparison. The pipeline is released as the `data_analysis.ipynb` notebook with the supporting feature-engineering module `feature_engineering.py`.
2. **Head-to-head benchmark.** Direct comparison of four generators spanning the GAN/LLM and DP/non-DP axes: HealthGAN [5], PATE-GAN [6], Pythia-LoRA [7, 8, 3], and Pythia-DP-LoRA (Opacus DP-SGD [9, 10]). The Pythia-DP variant constitutes a scope extension beyond the originally submitted proposal, yielding the first within-family DP-cost comparison for tabular LLM synthesis on this benchmark.
3. **Within-family DP cost characterisation.** Quantitative evidence on the marginal cost of formal ($\varepsilon = 5.0, \delta = 10^{-5}$)-differential privacy, separately measured for the GAN family (HealthGAN $\rightarrow$ PATE-GAN) and the LLM family (Pythia $\rightarrow$ Pythia-DP).
4. **Practical guidance.** A decision-oriented summary identifying when a non-DP GAN suffices, when formal DP via PATE-GAN is warranted, and where LLM-based synthesis remains competitive given privacy budgets and governance constraints.

8.2 Direct Answers to the Research Questions

RQ1 – Evaluation Methods.

RQ2 – Trade-offs Per Generator.

RQ3 – The Better Balance.

8.3 Future Work

8.3.1 Larger Language Models

8.3.2 Cross-Dataset Generalisation

8.3.3 Privacy-Budget Ablation

8.3.4 Stronger Privacy Adversaries

8.3.5 Longitudinal and Multimodal Records

8.4 Closing Remarks

Through a combined assessment of utility — spanning both distributional similarity to the real data and predictive performance of downstream models — alongside instance-level privacy risks, this thesis aims to determine not only which models perform best, but under which conditions each class of models is most appropriate. This enables a nuanced interpretation of when non-DP synthesis is sufficient, when formal differential privacy is necessary, and how LLM-based generators behave in terms of memorisation and disclosure risk in clinical settings. Beyond generating comparative insights, the thesis contributes a reusable, methodologically grounded evaluation protocol that can support future research and practical governance. Although the study focuses on tabular EHR-like data and a defined set of generative models, the framework provides a foundation for subsequent extensions to longitudinal records, multimodal data, and emerging DP-enhanced LLMs. Ultimately, the findings are intended to inform model selection, guide responsible data-release strategies, and strengthen the methodological basis for safe and high-utility synthetic data use in healthcare research.

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Appendix A

Hyperparameter Tables

A.1 HealthGAN (WGAN-GP)

A.2 PATE-GAN

A.3 Pythia (LoRA)

A.4 Pythia-DP (LoRA + Opacus)

Appendix B

Per-Column Distribution Plots

Appendix C

Per-Attribute Privacy Risk

Appendix D

Reproducibility

D.1 Repository Layout

D.2 Run Commands

D.3 Run Metadata Excerpts

D.4 Computational Environment

Appendix E

HealthGAN Variant Comparison